Vicente Kabeer Pedreros Aballay

Personal information

- **Full name**Vicente Kabeer Pedreros Aballay
- **Place of birth**Rancagua, Región del Libertador Bernardo O'Higgins, Chile.
- **E-Mail**vicente dot pedreros at ug dot uchile dot cl

Languages

- **Spanish**Native.
- **English**Advanced (TOEFL iBT 105/120, taken on Oct 2025, 110/120 My Best Scores).

Formal Education

- **Facultad de ciencias Físicas y Matemáticas, Universidad de Chile** Santiago, Chile
  Master's degree in Physics *Aug 2023 - Ongoing*
- **Facultad de ciencias Físicas y Matemáticas, Universidad de Chile** Santiago, Chile
  Bachelor's degree in Astronomy, Outstanding student *Mar 2019 - Jul 2023*

Research Experience

- **Physics department, FCFM, Universidad de Chile** Santiago, Chile
  Collaboration with Prof. Cristian Barrera (Universidad de Valparaíso) *Dec 2024- Ongoing*
  - **Project:**Quantifying the signal-to-noise ratio of the cross-correlation between the weak-lensing convergence field and the kinetic Sunyaev-Zeldovich effect in models beyond ΛCDM.
- **Physics department, FCFM, Universidad de Chile** Santiago, Chile
  Master's Thesis *Jan 2024- Ongoing*
  - **Title:**Machine Learning-Based Emulators for the Non-Linear Matter Power Spectrum in Current Cosmological Surveys.
- **Physics department, FCFM, Universidad de Chile** Santiago, Chile
  Summer Internship *Jan 2023*
  - **Activities:**Collaborative work on predicting cosmological parameters using the weak lensing observable, mainly using `Python` and `CAMB`. Improvement in 2 orders of magnitude on the errors.
- **Astronomy department, FCFM, Universidad de Chile** Las Condes, Chile
  Research Project on Astronomy *Sep 2022 - Dec 2022*
  - **Activities:**Mass segregation study on *Coma Berenices* and *Ursa Major II*, using `Python`. No evidence for mass segregation was found, agreeing with the current categorization of the objects.
- **Astronomy department, FCFM, Universidad de Chile** Las Condes, Chile
  Summer Internship *Jan 2022*
  - **Activities:**Collaborative work on classifier of gravitational lenses of ALeRCE.

Collaborations

- **LSST Dark Energy Science Collaboration**
  Higher Order Statistics topical team: building mass maps for LSST *Jan 2025 - Ongoing*

Teaching Experience

- **Physics department, FCFM, Universidad de Chile** Santiago, Chile
  Mar 2023 - Aug 2025
  - Teaching Assistant for Electromagnetism, and graduate level Electrodynamics and Cosmology. Grader for General Relativity.

Schools and Workshops

- **Michigan Cosmology Summer School** Michigan, US
  Jun 2025
- **Workshop: from Curauma to the Universe** Valparaíso, Chile
  Participated as a speaker: On EFTofLSS and how to include dark energy perturbations. *20 Aug, 2024*
- **Summer School on Cosmology** Trieste, Italy
  Jun 2024
- **Pacific Summer School** Santiago, Chile
  Jan 2024

Outreach

- **Talk at Public School** San Bernardo, Chile
  Liceo Nuevo Porvenir *14 Sep, 2023*
- **Rubin's First Look at FCFM** Santiago, Chile
  Main organizer and delivered the introductory talk *21 Jun, 2025*

Computer Skills

- `Python`Proficient.
- **LaTeX**Proficient.
- **Cosmology**`CAMB` and `CLASS`.
- **Operating Systems**Linux, OSX and Windows.
- **Misc.**Familiar with Vim, git and shell.